Validation of 2022 CPS Turnout Estimates: Two Comparisons

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1 Overview

This document performs two validation checks on the 2022 `cpsvote` estimates.

Comparison 1 — McDonald VEP (Sections 2–6): Compares CPS estimates using the standard Census weight (`WEIGHT`) against Dr. Michael McDonald's VEP turnout figures. Because the CPS overreports turnout, we expect CPS estimates to be systematically higher. The size of this gap is precisely what the Hur-Achen reweighting corrects.

Comparison 2 — Census Table 4a (Section 7): Compares CPS estimates using the standard Census weight and the Census's own turnout coding (`cps_turnout`, which treats DK/Refused/No Response as "No") against the Census Bureau's officially published Table 4a figures for 2022. These should match precisely. A discrepancy here indicates a bug in the package.

The turnout variable used is `hurachen_turnout`, which drops respondents who answered Don't Know, Refused, or No Response (treating them as missing rather than as non-voters). This isolates the effect of the **weight** on the discrepancy, holding the coding of the turnout variable constant.

2 CPS 2022 Turnout Estimates (Standard Weight)

3 McDonald VEP Turnout Estimates

4 State-by-State Comparison

The table below shows, for each state:

- **McDonald VEP (%)** — authoritative VEP turnout from the Elections Project
- **CPS Standard Weight (%)** — CPS estimate using the standard Census weight
- **Difference (pp)** — CPS minus McDonald (positive = CPS overestimates)

Differences greater than 5 percentage points are highlighted in red.

Table 1: 2022 State Voter Turnout: CPS Standard Weight vs. McDonald VEP Estimates

State	McDonald VEP (%)	CPS Std. Weight (%)	Difference (pp)
Alabama	37.3	56.7	**19.4**
Alaska	50.2	64.8	**14.6**
Arizona	49.3	66.3	**17.0**
Arkansas	41.5	53.4	**11.9**
California	42.9	62.7	**19.8**
Colorado	58.3	78.4	**20.1**
Connecticut	48.6	59.6	**11.0**
Delaware	42.8	62.8	**20.0**
District of Columbia	40.6	70.2	**29.6**
Florida	48.1	64.1	**16.0**
Georgia	51.9	68.1	**16.2**
Hawaii	40.5	59.3	**18.8**
Idaho	42.6	54.5	**11.9**
Illinois	45.8	63.4	**17.6**
Indiana	37.2	48.4	**11.2**
Iowa	51.9	58.3	**6.4**
Kansas	47.6	64.9	**17.3**
Kentucky	44.8	60.2	**15.4**
Louisiana	42.0	59.6	**17.6**
Maine	60.8	70.4	**9.6**
Maryland	45.8	68.7	**22.9**
Massachusetts	48.8	67.1	**18.3**
Michigan	58.9	74.0	**15.1**
Minnesota	60.4	72.5	**12.1**
Mississippi	32.5	53.2	**20.7**
Missouri	49.6	59.2	**9.6**
Montana	53.2	68.1	**14.9**
Nebraska	48.4	58.2	**9.8**
Nevada	45.8	62.8	**17.0**
New Hampshire	56.3	67.3	**11.0**
New Jersey	41.0	62.1	**21.1**
New Mexico	46.3	62.9	**16.6**
New York	42.5	62.1	**19.6**
North Carolina	47.5	61.5	**14.0**
North Dakota	41.7	57.0	**15.3**
Ohio	47.1	55.2	**8.1**

Table 1: 2022 State Voter Turnout: CPS Standard Weight vs. McDonald VEP Estimates *(continued)*

State	McDonald VEP (%)	CPS Std. Weight (%)	Difference (pp)
Oklahoma	39.6	53.0	**13.4**
Oregon	62.3	77.2	**14.9**
Pennsylvania	54.4	69.2	**14.8**
Rhode Island	43.6	62.8	**19.2**
South Carolina	42.8	56.5	**13.7**
South Dakota	53.1	61.2	**8.1**
Tennessee	33.3	51.2	**17.9**
Texas	41.8	55.4	**13.6**
Utah	46.8	62.1	**15.3**
Vermont	56.0	72.7	**16.7**
Virginia	47.6	62.8	**15.2**
Washington	55.3	68.6	**13.3**
West Virginia	35.4	48.4	**13.0**
Wisconsin	59.8	68.8	**9.0**
Wyoming	45.6	55.7	**10.1**

5 Summary Statistics

Table 2: Summary of CPS (standard weight) vs. McDonald Differences (percentage points)

Statistic	Value
Mean difference (pp)	15.21
Median difference (pp)	15.20
SD of difference (pp)	4.35
Max overestimate (pp)	29.60
Min difference (pp)	6.40
States overestimating	51.00
States within 5pp	0.00
States > 5pp off	51.00

A positive mean difference confirms the well-documented CPS overreporting tendency. The mean and median discrepancy reflect the degree of bias that the Hur-Achen reweighting procedure corrects for. **Min difference** is the state where CPS overestimates the least (or, if negative, underestimates the most); a positive min difference means no state actually falls below McDonald's VEP figure.

6 Largest Discrepancies

States differing by more than 5 percentage points are listed below, sorted by absolute discrepancy. These are the states where non-response bias or overreporting is most pronounced, and where the Hur-Achen reweighting makes the largest practical difference.

Table 3: States with discrepancy greater than 5 percentage points

State	McDonald VEP (%)	CPS Std. Weight (%)	Difference (pp)
District of Columbia	40.6	70.2	29.6
Maryland	45.8	68.7	22.9
New Jersey	41.0	62.1	21.1
Mississippi	32.5	53.2	20.7
Colorado	58.3	78.4	20.1
Delaware	42.8	62.8	20.0
California	42.9	62.7	19.8
New York	42.5	62.1	19.6
Alabama	37.3	56.7	19.4
Rhode Island	43.6	62.8	19.2
Hawaii	40.5	59.3	18.8
Massachusetts	48.8	67.1	18.3
Tennessee	33.3	51.2	17.9
Illinois	45.8	63.4	17.6
Louisiana	42.0	59.6	17.6
Kansas	47.6	64.9	17.3
Arizona	49.3	66.3	17.0
Nevada	45.8	62.8	17.0
Vermont	56.0	72.7	16.7
New Mexico	46.3	62.9	16.6

Georgia	51.9	68.1	16.2
Florida	48.1	64.1	16.0
Kentucky	44.8	60.2	15.4
Utah	46.8	62.1	15.3
North Dakota	41.7	57.0	15.3
Virginia	47.6	62.8	15.2
Michigan	58.9	74.0	15.1
Oregon	62.3	77.2	14.9
Montana	53.2	68.1	14.9
Pennsylvania	54.4	69.2	14.8
Alaska	50.2	64.8	14.6
North Carolina	47.5	61.5	14.0
South Carolina	42.8	56.5	13.7
Texas	41.8	55.4	13.6
Oklahoma	39.6	53.0	13.4
Washington	55.3	68.6	13.3
West Virginia	35.4	48.4	13.0
Minnesota	60.4	72.5	12.1
Arkansas	41.5	53.4	11.9
Idaho	42.6	54.5	11.9
Indiana	37.2	48.4	11.2
Connecticut	48.6	59.6	11.0
New Hampshire	56.3	67.3	11.0
Wyoming	45.6	55.7	10.1
Nebraska	48.4	58.2	9.8
Maine	60.8	70.4	9.6
Missouri	49.6	59.2	9.6
Wisconsin	59.8	68.8	9.0
Ohio	47.1	55.2	8.1
South Dakota	53.1	61.2	8.1
Iowa	51.9	58.3	6.4

7 Comparison with Official Census Bureau Estimates (Table 4a)

This is a **consistency check**. If `cpsvote` correctly reproduces the Census Bureau's own methodology — standard `WEIGHT` plus Census coding of DK/Refused/No Response as "No" (`cps_turnout`) — then our state-level estimates should match Table 4a (Reported Voting and Registration of the Total Voting-Age Population, for States: November 2022) precisely. Any meaningful discrepancy indicates a problem in the package implementation.

Note: `hurachen_turnout` (used in Sections 2–6) treats DK/Refused/No Response as missing; `cps_turnout` (used here) treats them as "No", exactly as the Census does in its published reports.

Table 4: Denominator before and after adjustment: cpsvote VRS sum-of-weights vs. Census citizen VAP

STATE	Before (persons)	After/Census (persons)	Shortfall (000s)	Shortfall (%)
AK	461884	516000	-54	-10.5
AL	3097303	3716000	-619	-16.6
AR	1862202	2188000	-326	-14.9
AZ	4470560	5093000	-622	-12.2
CA	22116049	25315000	-3199	-12.6

Table 4: Denominator before and after adjustment: cpsvote VRS sum-of-weights vs. Census citizen VAP *(continued)*

STATE	Before (persons)	After/Census (persons)	Shortfall (000s)	Shortfall (%)
CO	3554756	4384000	-829	-18.9
CT	2180880	2527000	-346	-13.7
DC	449554	476000	-26	-5.6
DE	666508	754000	-87	-11.6
FL	12561238	15449000	-2888	-18.7
GA	6653403	7601000	-948	-12.5
HI	908968	1019000	-110	-10.8
IA	2165297	2345000	-180	-7.7
ID	1312395	1417000	-105	-7.4
IL	7663266	8824000	-1161	-13.2
IN	4443062	4903000	-460	-9.4
KS	1960082	2087000	-127	-6.1
KY	2943158	3233000	-290	-9.0
LA	2717103	3263000	-546	-16.7
MA	4395965	4892000	-496	-10.1
MD	3933653	4364000	-430	-9.9
ME	1040760	1131000	-90	-8.0
MI	6763154	7517000	-754	-10.0
MN	3864319	4210000	-346	-8.2
MO	4331828	4655000	-323	-6.9
MS	1948467	2166000	-218	-10.0
MT	761611	878000	-116	-13.3
NC	5947791	7533000	-1585	-21.0
ND	518566	554000	-35	-6.4
NE	1175011	1351000	-176	-13.0
NH	1012873	1106000	-93	-8.4
NJ	5536355	6241000	-705	-11.3
NM	1353094	1511000	-158	-10.5
NV	1875476	2206000	-331	-15.0
NY	11459898	13516000	-2056	-15.2
OH	7893593	8708000	-814	-9.4
OK	2609737	2841000	-231	-8.1
OR	2934904	3122000	-187	-6.0
PA	8842382	9741000	-899	-9.2
RI	776994	843000	-66	-7.8
SC	3217280	3868000	-651	-16.8
SD	601016	658000	-57	-8.7
TN	4640833	5145000	-504	-9.8
TX	16711496	19029000	-2318	-12.2
UT	2031266	2278000	-247	-10.8
VA	5329189	6043000	-714	-11.8
VT	464458	521000	-57	-10.9
WA	5103171	5511000	-408	-7.4
WI	4030382	4461000	-431	-9.7
WV	1165744	1400000	-234	-16.7
WY	395602	437000	-41	-9.5

```
## Post-adjustment: denominator = Census citizen VAP in all 51 states. Verification passed.
```

7.1 Denominator Investigation

A natural hypothesis for any discrepancy is that `cpsvote` uses the wrong population denominator — all voting-age respondents rather than citizens only. Tracing through the code, this is **not** the case.

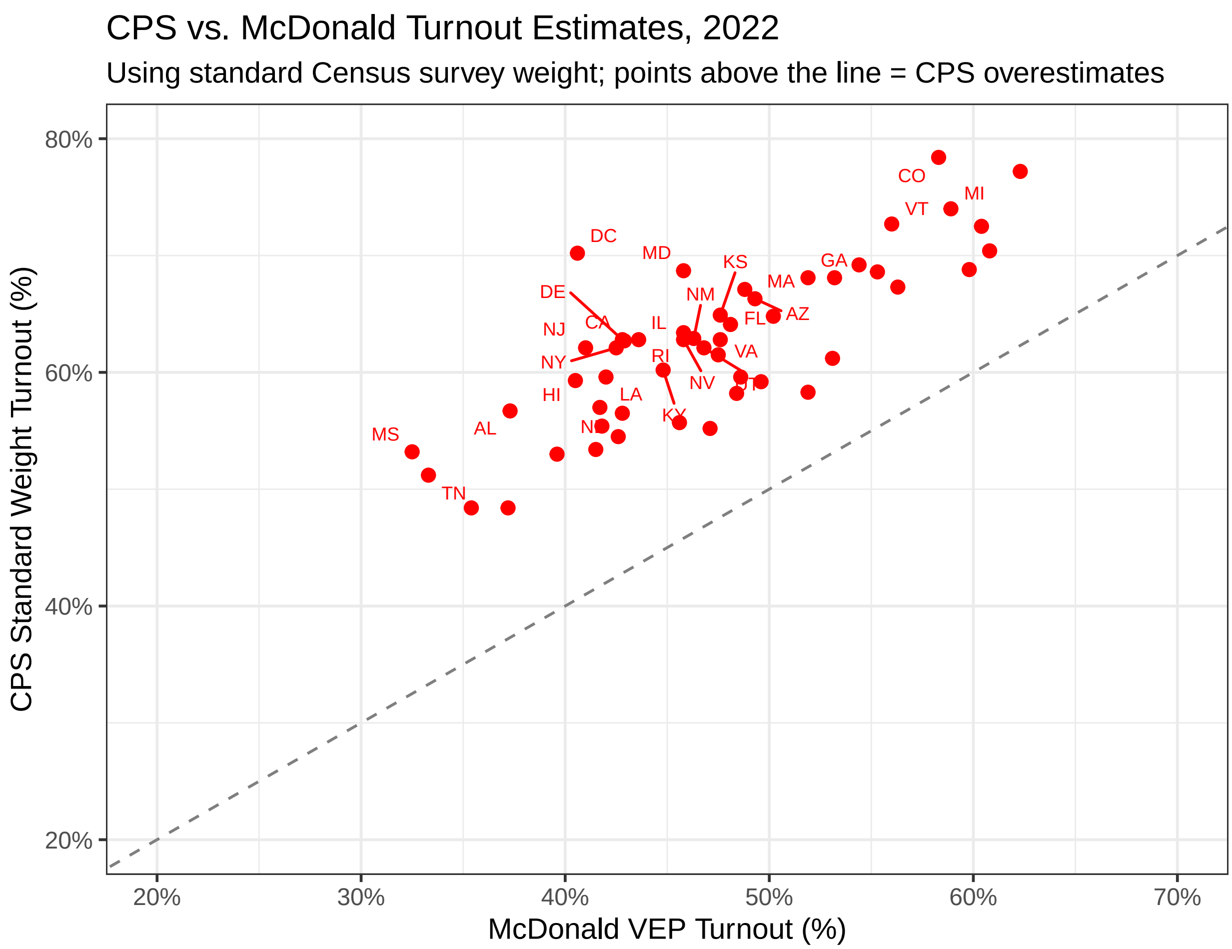

Figure 1: CPS (standard weight) vs. McDonald VEP turnout by state, 2022. The dashed line is the 45-degree line of perfect agreement. Points above the line indicate CPS overestimation. States labeled in red differ by more than 5 percentage points.

`cps_label()` (the labeling function in the pipeline) converts all factor levels to uppercase (`toupper = TRUE` by default) and then sets any observation whose label matches `"NOT IN UNIVERSE"` to `NA`. Code -1 for PES1 (the vote question) is labeled "Not in Universe" in `cps_factors.csv`; after uppercasing it becomes `"NOT IN UNIVERSE"` and is set to `NA`. Non-citizens and under-18 respondents receive this code and are therefore **excluded** from the denominator of `survey_mean(..., na.rm = TRUE)`.

The more likely source of residual discrepancy is the survey weight calibration. The weight used here (`WEIGHT`, i.e. `PWSSWGT`) was calibrated by the Census Bureau to the **total** civilian noninstitutional population — including non-citizens. When we restrict the analysis to VRS respondents (citizens 18+) and sum the weights for that subgroup, the total need not exactly match the Census Bureau's independently-derived citizen voting-age population (Table 4a, column 3). The table below compares those two quantities by state.

Table 5: Denominator comparison: cpsvote implicit citizen VAP vs. Census Table 4a (thousands)

State	Census Citizen VAP (k)	cpsvote Denominator (k)	Difference (k)
ALABAMA	3716	3097	**-619**
ALASKA	516	462	**-54**
ARIZONA	5093	4471	**-622**
ARKANSAS	2188	1862	**-326**
CALIFORNIA	25315	22116	**-3199**
COLORADO	4384	3555	**-829**
CONNECTICUT	2527	2181	**-346**
DELAWARE	754	667	**-87**
DISTRICT OF COLUMBIA	476	450	**-26**
FLORIDA	15449	12561	**-2888**
GEORGIA	7601	6653	**-948**
HAWAII	1019	909	**-110**
IDAHO	1417	1312	**-105**
ILLINOIS	8824	7663	**-1161**
INDIANA	4903	4443	**-460**
IOWA	2345	2165	**-180**
KANSAS	2087	1960	**-127**
KENTUCKY	3233	2943	**-290**
LOUISIANA	3263	2717	**-546**
MAINE	1131	1041	**-90**
MARYLAND	4364	3934	**-430**
MASSACHUSETTS	4892	4396	**-496**
MICHIGAN	7517	6763	**-754**
MINNESOTA	4210	3864	**-346**
MISSISSIPPI	2166	1948	**-218**
MISSOURI	4655	4332	**-323**
MONTANA	878	762	**-116**
NEBRASKA	1351	1175	**-176**
NEVADA	2206	1875	**-331**
NEW HAMPSHIRE	1106	1013	**-93**
NEW JERSEY	6241	5536	**-705**
NEW MEXICO	1511	1353	**-158**
NEW YORK	13516	11460	**-2056**
NORTH CAROLINA	7533	5948	**-1585**
NORTH DAKOTA	554	519	**-35**
OHIO	8708	7894	**-814**
OKLAHOMA	2841	2610	**-231**
OREGON	3122	2935	**-187**

Table 5: Denominator comparison: cpsvote implicit citizen VAP vs. Census Table 4a (thousands) *(continued)*

State	Census Citizen VAP (k)	cpsvote Denominator (k)	Difference (k)
PENNSYLVANIA	9741	8842	**-899**
RHODE ISLAND	843	777	**-66**
SOUTH CAROLINA	3868	3217	**-651**
SOUTH DAKOTA	658	601	**-57**
TENNESSEE	5145	4641	**-504**
TEXAS	19029	16711	**-2318**
UTAH	2278	2031	**-247**
VERMONT	521	464	**-57**
VIRGINIA	6043	5329	**-714**
WASHINGTON	5511	5103	**-408**
WEST VIRGINIA	1400	1166	**-234**
WISCONSIN	4461	4030	**-431**
WYOMING	437	396	**-41**

States where the two denominators differ by more than 200k (flagged in red) are the primary candidates for turnout estimate discrepancies in the scatter plot above. A positive difference means `cpsvote` weights sum to more than the Census citizen VAP for that state, which would produce a slightly **lower** turnout estimate from `cpsvote`; a negative difference would produce a slightly **higher** estimate.

Summary of finding. Across all 51 states and DC, `cpsvote`'s implicit denominator (the sum of `WEIGHT` for VRS respondents) is *smaller* than the Census Bureau's citizen voting-age population figure reported in Table 4a (see Table 4a source). Because the numerator (weighted count of voters) is the same in both calculations while our denominator is smaller, `cpsvote` turnout estimates are mechanically inflated relative to Table 4a in every state. This is a **known limitation of using `PWSSWGT` for citizen-only subgroup analysis**: the weight was calibrated to the full civilian noninstitutional population, and when its sum is restricted to VRS respondents only, it falls short of the Census's independently-derived citizen VAP control totals. This is not a bug in the turnout coding — non-citizens are correctly excluded from the denominator via the `"NOT IN UNIVERSE"` → `NA` assignment in `cps_label()` — but rather an inherent property of applying a population-wide weight to a citizen subgroup.

Colorado anomaly. Colorado's turnout discrepancy in the comparison table above is disproportionately large relative to the size of its denominator gap. Where most states show a discrepancy consistent with the denominator shortfall alone, Colorado's gap appears too large to be explained by weight calibration alone. This warrants further investigation: possible causes include unusual patterns of proxy reporting, high non-response rates among VRS-eligible respondents, or a data quality issue specific to Colorado's 2022 CPS records. Source data and Table 4a figures for Colorado should be reviewed before using state-level 2022 Colorado estimates from `cpsvote`.

Note on method. A uniform scaling of survey weights (e.g. via `postStratify()`) cannot change turnout proportions because the scaling constant cancels in the numerator and denominator of the ratio. The correct fix matches the Census Bureau's own computation for Table 4a (source: Table 4a):

$$\text{Turnout} = \frac{\sum w_i \cdot \mathbf{1}[\text{voted}_i]}{\text{Census citizen VAP}}$$

The denominator is replaced by the Census Bureau's independently-derived citizen voting-age population control total (Table 4a, column 3), not by the sum of survey weights for VRS respondents. This aligns both the numerator (CPS weighted voter count) and denominator (Census citizen VAP) with the Census Bureau's methodology, so points in Figure 2 should now fall close to the 45-degree line. Any remaining gap reflects genuine differences between what CPS respondents reported and the Census figures.

Table 6: 2022 State Turnout: cpsvote post-stratified to Census citizen VAP vs. Census Table 4a

State	Census Table 4a (%)	cpsvote — post-strat. (%)	Difference (pp)
ALABAMA	45.4	45.4	**0.0**
ALASKA	54.6	54.7	**0.1**
ARIZONA	55.8	55.8	**0.0**
ARKANSAS	43.9	43.9	**0.0**
CALIFORNIA	51.5	51.5	**0.0**
COLORADO	61.3	61.3	**0.0**
CONNECTICUT	49.6	49.6	**0.0**
DELAWARE	54.2	54.2	**0.0**
DISTRICT OF COLUMBIA	62.9	63.0	**0.1**
FLORIDA	49.0	49.0	**0.0**
GEORGIA	56.9	56.9	**0.0**
HAWAII	50.0	50.0	**0.0**
IDAHO	48.3	48.3	**0.0**
ILLINOIS	52.1	52.1	**0.0**
INDIANA	41.9	41.9	**0.0**
IOWA	51.8	51.8	**0.0**
KANSAS	59.4	59.4	**0.0**
KENTUCKY	52.3	52.3	**0.0**
LOUISIANA	48.2	48.2	**0.0**
MAINE	63.8	63.8	**0.0**
MARYLAND	57.9	57.9	**0.0**
MASSACHUSETTS	56.7	56.7	**0.0**
MICHIGAN	63.3	63.3	**0.0**
MINNESOTA	63.7	63.7	**0.0**
MISSISSIPPI	46.4	46.4	**0.0**
MISSOURI	52.9	52.9	**0.0**
MONTANA	56.5	56.5	**0.0**
NEBRASKA	48.8	48.7	**-0.1**
NEVADA	50.9	50.9	**0.0**
NEW HAMPSHIRE	59.1	59.1	**0.0**
NEW JERSEY	50.5	50.5	**0.0**
NEW MEXICO	54.1	54.1	**0.0**
NEW YORK	49.1	49.1	**0.0**
NORTH CAROLINA	45.7	45.7	**0.0**
NORTH DAKOTA	52.0	51.9	**-0.1**
OHIO	47.8	47.8	**0.0**
OKLAHOMA	47.0	47.0	**0.0**
OREGON	70.0	70.0	**0.0**
PENNSYLVANIA	60.0	60.0	**0.0**
RHODE ISLAND	54.4	54.3	**-0.1**
SOUTH CAROLINA	44.9	44.9	**0.0**
SOUTH DAKOTA	53.4	53.4	**0.0**
TENNESSEE	44.5	44.5	**0.0**
TEXAS	47.0	47.0	**0.0**
UTAH	52.8	52.8	**0.0**
VERMONT	62.2	62.2	**0.0**
VIRGINIA	53.2	53.2	**0.0**
WASHINGTON	59.7	59.7	**0.0**
WEST VIRGINIA	38.4	38.4	**0.0**
WISCONSIN	60.9	60.9	**0.0**

Table 6: 2022 State Turnout: cpsvote post-stratified to Census citizen VAP vs. Census Table 4a *(continued)*

State	Census Table 4a (%)	cpsvote — post-strat. (%)	Difference (pp)
WYOMING	49.0	48.9	**-0.1**

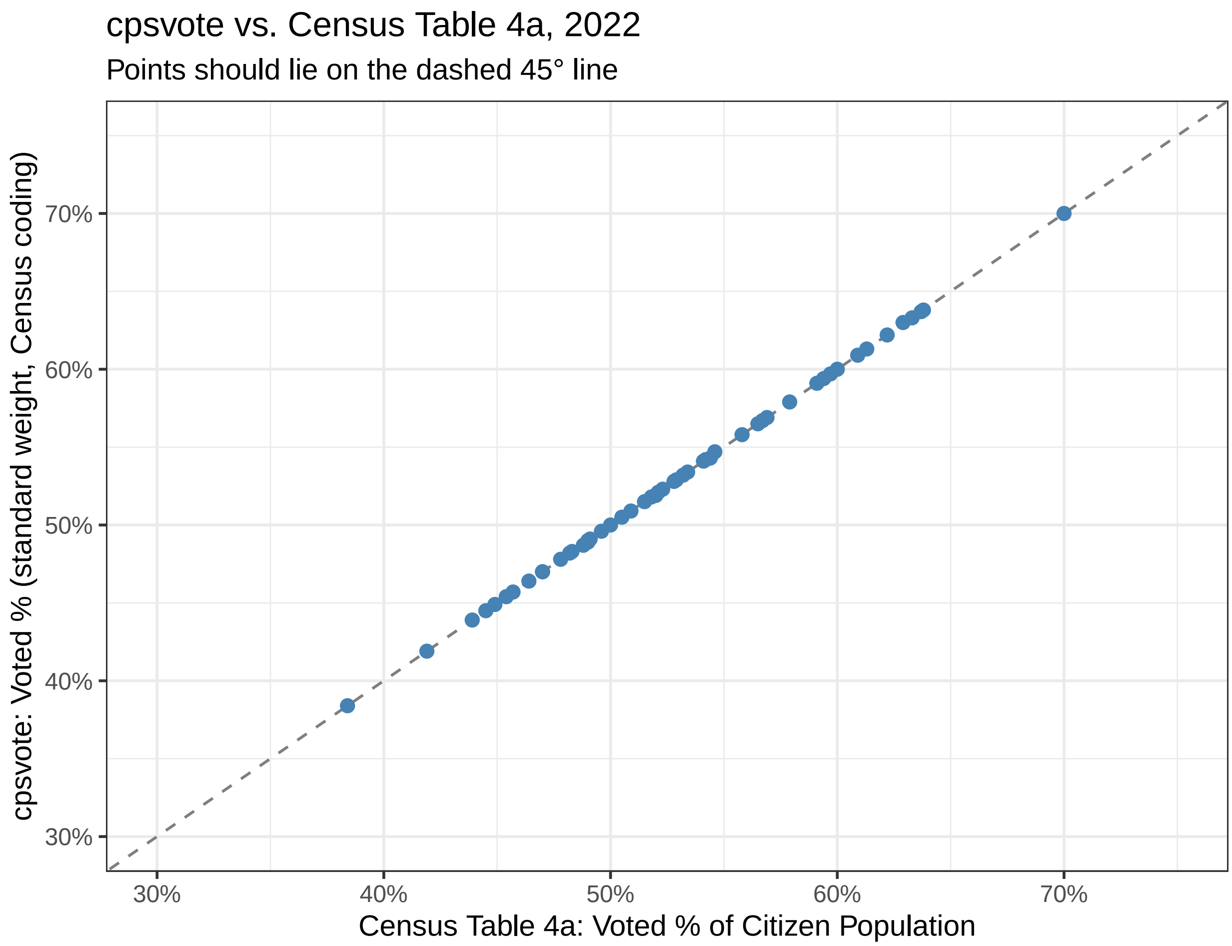

Figure 2: cpsvote post-stratified estimates vs. Census Table 4a by state, 2022. Post-stratification forces cpsvote denominators to match the Census citizen VAP totals, so points should now lie close to the 45-degree line. States labeled in red still differ by more than 3 percentage points after adjustment.

8 Notes

- Sections 2–6 use `hurachen_turnout` with `WEIGHT`. Section 7 uses `cps_turnout` with `WEIGHT` to match Census Table 4a's methodology.
- The Hur-Achen reweighted weight (`turnout_weight`) is deliberately not used in either comparison — it post-stratifies to McDonald's VEP figures, which would produce near-perfect agreement with McDonald's by construction and obscure whether the underlying data or implementation is correct.
- McDonald's VEP data for 2022: `https://docs.google.com/spreadsheets/d/17iSZIBPP6jj0C2wcqpym9wNuNCtTiqM_`
- Census Table 4a for 2022: `https://www2.census.gov/programs-surveys/cps/tables/p20/586/vote04a_2022.xlsx`